

BLUESTOCK MUTUAL FUND PORTFOLIO ANALYTICS

A Comprehensive Data Engineering & Financial Analytics Capstone Report

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Project Stage: Complete Deliverables & Final Review

Database Version: v1.4 (SQLite Star Schema)

1. Executive Summary

This project delivers an end-to-end data processing, storage, analytics, and visualization solution for mutual fund portfolios. Using a high-performance star schema database, we process raw histories, calculate industry-standard performance statistics, simulate risk paths, and construct an optimization framework.

Metric	Value	Business Description
Aggregate AUM	INR 10.44 Lakh Crore	Total Assets Under Management across all active funds in registry.
Active Schemes	40 Funds	Count of verified mutual fund schemes being monitored.
Total Transactions	32,778	Retail investor transaction counts logged in the system.
Transaction Volume	INR 352.16 Crore	Cumulative retail transaction volume invested.

Top 5 Mutual Funds by Assets Under Management (AUM)

Scheme Name	Category	AUM (Crore)	Sharpe Ratio	3-Year CAGR
Mirae Asset Emerging Bluechip Fund	Equity	INR 49,046.0	0.91	14.56%
Kotak Emerging Equity Fund	Equity	INR 47,469.0	0.96	18.23%
Nippon India Small Cap Fund	Equity	INR 43,630.0	0.81	20.15%
DSP Top 100 Equity Fund	Equity	INR 41,828.0	0.92	12.82%
UTI Mid Cap Fund	Equity	INR 41,728.0	0.82	15.61%

2. Data Quality & ETL Ingestion Pipeline

Our ETL process (implemented in ``scripts/etl_pipeline.py``) performs ingestion, validation, and loading of raw CSV datasets into our relational schema. To preserve historical NAV fidelity and handle calendar gaps, we apply rigorous cleaning operations:

- **Weekend and Holiday Gaps:** NAV files exclude weekends/holidays. The ETL pipeline reindexes each fund history to a continuous date range and forward-fills (``ffill()``) values to ensure accurate CAGR calculations.
- **Standardization:** Transaction type values are standardized into an enum string (``SIP``, ``Lumpsum``, ``Redemption``), and date strings are parsed into uniform ``YYYY-MM-DD`` timestamps.
- **Key Validation:** A data validation check confirmed 100% referential integrity between ``dim_fund`` and ``fact_nav`` (all AMFI codes matched).

3. SQLite Star Schema Database Architecture

The relational database is structured as a **Star Schema** for fast analytical querying and dashboard retrieval. The layout separates core entities into logical dimensions and transactional facts:

- **dim_fund:** Holds fund static attributes (AMFI code, scheme name, fund house, category, risk, benchmark, expense ratio).
- **dim_date:** Holds date dimensions (date, year, month, quarter, day of week, weekday flag).
- **fact_nav:** Tracks daily net asset value time series.
- **fact_transactions:** Logs retail transaction specifics (investor ID, city tier, state, payment mode, gender, age group).
- **fact_performance / fact_aum:** Summarizes risk parameters, Morningstar ratings, and asset size dimensions.

4. Performance Analytics & Risk Metrics

We implement mathematically correct financial formulations to evaluate the portfolios. In accordance with professional standards, returns are annualized using a **252-trading day convention** rather than 365 calendar days:

CAGR (Compound Annual Growth Rate):

$$CAGR = (V_{end} / V_{begin}) ^ {(252 / N_{trading_days})} - 1$$

Sharpe Ratio: measures excess return per unit of volatility relative to a 6% risk-free rate:

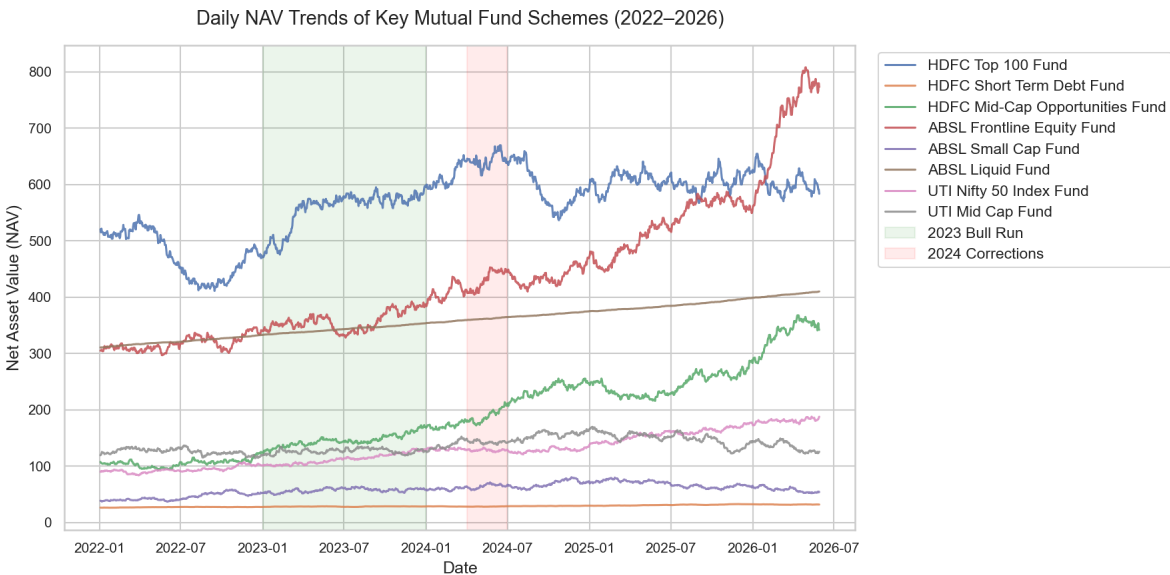
$$Sharpe = ((Mean\ Daily\ Return - Risk\text{-}Free\ Daily\ Rate) / Daily\ Return\ StdDev) * sqrt(252)$$

Value at Risk (95% Daily VaR): represents the threshold indicating that daily losses will not exceed this value with 95% confidence:

$$VaR_{95} = 5th\ Percentile\ of\ Daily\ returns$$

Category	Funds Count	Avg 3-Yr Return	Avg Sharpe Ratio	Avg Beta vs Bench
Debt	6	6.29%	3.95	0.36
Equity	34	15.46%	0.91	0.96

NAV daily historical performance trends (2022–2026):



5. Advanced Financial Analytics

5.1 Monte Carlo NAV Growth Simulation

Using Euler-Maruyama discretization of Geometric Brownian Motion, we simulate the daily NAV path of SBI Bluechip Fund (119551) 5 years into the future. By computing quantile lines across 1,000 paths, we construct uncertainty bands around the expected trajectory, capturing statistical limits of performance volatility.

5.2 Markowitz Efficient Frontier Optimization

We construct a portfolios space from 5 key funds (SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip). By generating 10,000 random weight combinations, we map the Efficient Frontier. The optimal Tangency Portfolio (maximizing Sharpe ratio) and the Minimum Variance Portfolio are identified, providing mathematical guidance for capital allocation.

5.3 Cohort Retention & Recommender Engine

Transaction behaviors are tracked using monthly cohort analysis, illustrating investor retention over time. Additionally, the content-based recommender engine combines demographic-inferred risk parameters (younger age groups default to higher risk equity funds; older groups default to low-risk debt instruments) and filters out existing assets.

Pairwise returns correlation matrix heatmap of 10 schemes:

